

Feasibility Study of Thermal Energy Storage System

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1 Introduction

Industrial buildings often rely heavily on electric chillers to meet large cooling demands. However, running chillers during daytime when outdoor temperatures are higher and electricity prices peak can be both expensive and taxing on the electrical grid. One promising way to shift part of this load to off-peak hours is through thermal energy storage (TES), which decouples the timing of cooling production from actual cooling demand.

Among various TES options (such as chilled-water tanks and phase change materials), ice storage has distinct advantages for industries with limited space. By freezing water into ice during off-peak periods, typically at night when ambient conditions are cooler and power is cheaper, an ice-based system stores large amounts of cooling in a relatively small footprint. When daytime cooling is needed, the ice melts, supplying cold water (or glycol) near 0 °C directly to the facility. This not only takes advantage of lower nighttime prices but also reduces peak-demand charges by easing the chiller load during high-price hours.

In this study, an ice-based TES system is evaluated for an existing industrial chiller plant. Hourly cooling demand data over a representative week in early May were analyzed. Three scenarios were simulated in Excel, accounting for ice-maker efficiency, heat-exchanger losses, and realistic storage capacities. To ensure that results can be extended beyond this one-week window, a consistent state-of-charge boundary was enforced: the ice inventory at the end of the week matches the starting amount. This avoids artificial cost or energy savings caused by ending with unusually high or low ice stocks.

The primary objective is to determine whether adding an ice TES system reduces overall costs and to identify the conditions under which storage becomes economically attractive. By comparing energy flows, storage requirements, and lifetime costs over a 15-year horizon, this analysis seeks to provide clear guidance on whether an ice-based retrofit can improve both operational efficiency and economic performance for industrial cooling applications.

2 Objective of the study

This analysis aims to study whether the storage could enhance the cooling system from an efficiency and from an economical perspective. In order to implement thermal energy storage, it was decided to install an additional pipeline connected to the chillers, so that the cooling supply could be both supplied directly to the industry or to the storage through a three-way valve. An illustration of this system can be seen in Figure 1.

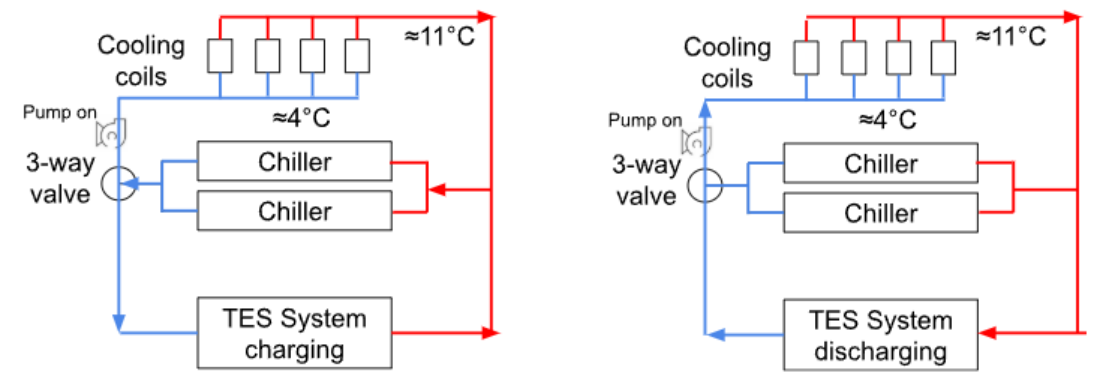


Figure 1: Scheme of the industry chiller plant

The first objective of this study is to minimize the peak thermal power demand by applying load shifting techniques, thereby distributing energy consumption more evenly over time and reducing strain on the energy system during peak periods. Second, to design and optimize a thermal energy storage (TES) system that is

both efficient and minimized in size, ensuring it can effectively support the shifted load while maintaining thermal performance and reducing material and installation costs. Finally, the project aims to study the cost-effectiveness of overall system configurations by integrating two different control strategies.

3 Overview of TES systems

There are several ways to implement TES in a cooling context. The main solutions include chilled water tanks, ice storage systems, and other phase change materials (PCMs).

The simplest method involves sensible heat storage through chilled water. In this setup, chillers cool water to around 4 to 6 °C, and the cold water is held in insulated tanks, usually designed to avoid the mixing of warmer return water from the cold supply. Even if simple and relatively cheap, this kind of system requires a lot of space; moreover, water does not store high quantity of energy per liter, meaning that high cooling demand requires storing a very large volume of liquid. For many industrial buildings, especially those with space limitations, this can become a severe obstacle.

This is why ice thermal storage offers an appealing alternative by freezing water, usually using a glycol-based coolant, to finally form ice when the electricity prices are low. Later, when cooling is needed, the ice is melted and absorbs heat from the building's return flow, sending out cold water (or glycol) around 0 °C. Because of the large amount of energy required to melt ice, this method stores much more cooling using less space. In fact, compared to chilled water storage, ice systems can achieve similar cooling effects with tanks that are up to ten times smaller [1]. Moreover, melting ice keeps the discharge temperature very stable, which is ideal for managing consistent cooling loads. The main downside is that the chiller has to operate at lower temperatures to freeze water, which is not as energy-efficient. However, this is mostly offset by running the system at night, when it is cooler outside and electricity is cheaper.

There are also other types of PCM-based storage that do not rely on water. These systems use engineered materials, such as certain salts or waxes, that melt at specific temperatures. The idea is to pick a phase change temperature that matches one cooling system's operating range. This procedure can be useful because it allows the chiller to work more efficiently and removes the need for antifreeze fluids. However, these alternative PCMs usually come with trade-offs. They are usually more expensive than water, and they sometimes suffer from performance degradation after repeated cycles. Their energy density is also often lower than that of ice. While they offer more design flexibility, they may not be the most practical choice for large-scale cooling needs [2].

4 Data Analysis

The study was decided to be addressed to an industry case, located in Sweden. Looking at the provided dataset, it can be seen that the cooling demand of the building is quite stable. During the week from 1st of May to the 6th of May, the week for which data has been provided, the cooling demand varies from 885 kWh to 1244 kWh, and averages around 1000 kWh. The supply temperature varies from around 4.5 to 5.3°C but averages around 4.7°C; while the return temperature is around $10.5 \pm 1^\circ\text{C}$. The flow rate of air varies from 121 to 198 m^3/s , averaging at 146 m^3/s .

The hourly cooling demand is shown in Figure 2.

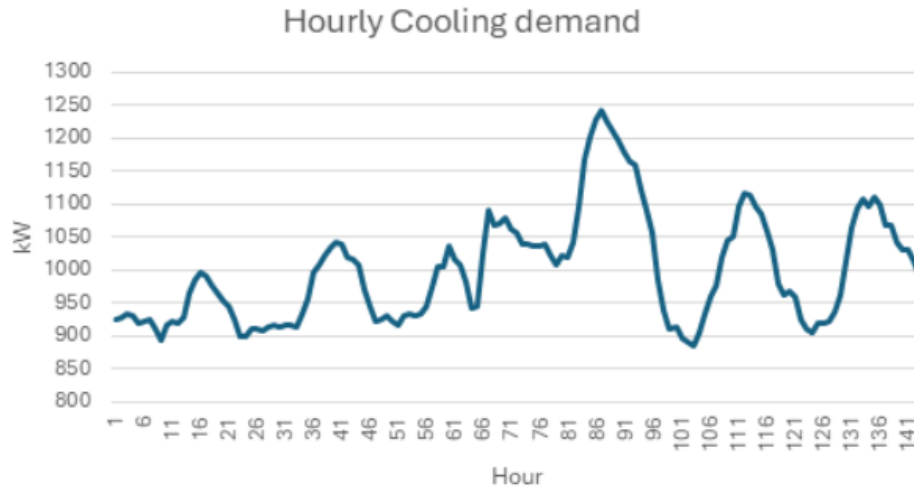


Figure 2: Industry's hourly cooling demand per week

Further studying the system, choosing ice as the phase change material for the TES is retained to be the best solution for this application. An ice storage system strikes a good balance between performance, reliability, and footprint. It conserves much cooling energy into a relatively compact form and support stable output temperatures. Moreover, they supply low temperatures, which are needed to meet a cooling demand of around 4-5°C. Nevertheless, it allows the facility to shift most of its cooling production to off-peak hours, taking advantage of lower electricity prices and reducing demand charges.

4.1 Missing Data

In the provided dataset, between 15h and 19h on 05/05, measured cooling demand, flow rate, supply and return temperature data were missing. To address this issue, interpolation was applied to estimate the missing values and ensure data continuity. It was observed that the data followed a downward trend during the afternoon, so the interpolation matched a descending pattern too. To achieve this, a simple linear interpolation method was applied. The difference between the preceding and following values was calculated and evenly divided into four parts, assuming a linear progression to estimate the missing data.

The calculations are shown in Table 1 and Figure 3.

Table 1: Data Values Over Time

Date				
05-05 13	1228.33	165	4.74	10.8
05-05 14	1244	186.83	5.28	10.48
05-05 15	1226.33	195.33	4.77	10.78
05-05 16	1211.17	191.75	4.77	10.66
05-05 17	1196.00	188.17	4.78	10.53
05-05 18	1180.84	184.58	4.78	10.41
05-05 19	1165.67	181	4.78	10.28
05-05 20	1159.67	180.83	4.73	10.17
05-05 21	1122	178.5	4.72	10.07

	A	B	C
93	05-05 15	1226.33	195.33
94	05-05 16	1211.17	191.75
95	05-05 17	1196.00	188.17
96	05-05 18	1180.84	184.58
97	05-05 19	1165.67	181

Figure 3: Data interpolation

5 Methodology

In order to understand when there will be some opportunities to profit from the installation of a storage system, 3 scenarios have been analyzed.

- Scenario 1 - Baseline with No Storage: this simple scenario served as a basis for comparison to the other two analysis.
- Scenario 2 - Storage with Constant Chiller Production: in this scenario the production of the chiller is maintained at a constant level without taking into consideration the hourly electricity prices.
- Scenario 3 - Storage with Variable Chiller Production: in this scenario the chiller production changes its output based on the electricity prices. Consequently, the chiller produces more cooling when the prices are low (filling the storage with the surplus) while lowering the production when the prices are higher (supplying the remainder cooling from the storage).

All simulations and calculation for the scenarios were performed using Excel.

A relevant consideration for the thermos-economic analysis of the scenarios that involve the storage, was the state of charge (SoC) at the beginning and at the end of the simulation period. The energy stored for the last hour of the dataset was set equal to the energy stored on the first hour, providing a technical assumption that ensures consistency with system operation, so that the work could be extended to longer periods (e.g., a year or the entire life of the storage). Without this boundary, any residual stored energy could generate a cost evaluation error, either overestimating profits or underestimating future energy needs.

6 Scenario 1: Baseline Scenario with no storage

As stated before, in this scenario the chiller's output follows exactly the demand curve. The economic calculations made were based on the dataset provided and the Swedish hourly prices [3]. Knowing the hourly cooling demand (kWh) and the COP of the chiller (Figure 6) the total weekly cost of the system has been calculated with the following formula:

$$Total\ Weekly\ Cost = \sum_{k=1}^N \frac{Cooling\ Demand_k * Hourly\ Price_k}{COP}$$

N is the total number of the hours considered in the dataset. The total weekly cost reaches 838.81€. In Figure 4 is also possible to see the different costs throughout the days while in Figure 5 the weekly price variance is represented.

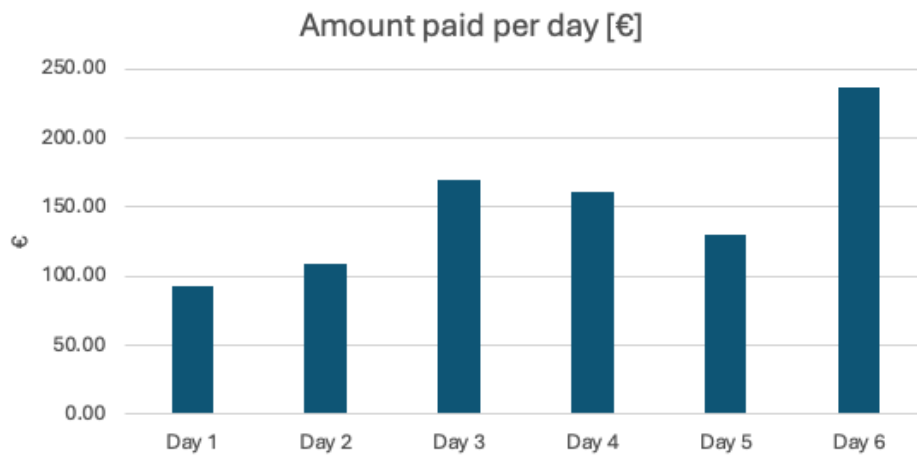


Figure 4: Amount paid per day



Figure 5: Hourly Prices throughout the week

Looking at Figures 4 and 5, it is clear that the total cost of the chiller's cooling production strongly depends on the electricity prices. Indeed, on the third and the sixth days, where the price is at its peak, the daily cost is much higher than for the other days.

It is also important to note that this variation does not depend on the chiller production since its level stands at almost at the same amount through all the week, with very low discrepancy (± 360 kWh).

From 5 is also possible to note that the hourly prices are very variable during the day, passing from very low prices (1-10 €/MWh) to high prices (50-70 €/MWh). This means that there is a much bigger variation in the prices than in the cooling demand, opening to different ways to exploit the energy storage scenarios. In Table 2 are the variation of the prices during the week is better exposed:

Table 2: Frequency distribution of hourly electricity prices

Hourly price (€/MWh)	Frequency (n° of hours in that range)
0–10	23
10–20	12
20–30	18
30–40	49
40–50	25
50–60	9
60–70	5
70–80	2

Extending the calculation to 15 years, which is the lifetime of the storage and chiller [4] [5], to make accurate comparison with the other scenarios can be made, the total cost accounts to around 1,133,000€, where the CAPEX for the two chillers is considered and the OPEX as well.

The operational costs are simply a multiplication of the total hours during the technical lifetime period of 15 years: having a dataset of 143 hours, to extend it in 131,400 hours, the OPEX of the dataset had to be multiplied by 918 (131,400 over 143 hours), to finally reach an operational expenditure of 775,345 € in the lifespan (the maintenance of the components is considered in their respective capital expenditures).

To estimate the chillers' CAPEX, on the other hand, the following formula was used:

$$CAPEX_{\text{chiller}} = \frac{\text{Nominal Capacity [kW]}}{3.516 \left[\frac{\text{kW}}{\text{ton}} \right]} \cdot 475 \left[\frac{\text{€}}{\text{ton}} \right]$$

The formula allows converting the nominal cooling capacity from kW to refrigeration tons using the standard

factor 1 ton = 3.516 kW and later multiplying the result by the average cost per ton.

The nominal power was selected based on the maximum cooling demand found, which reached 1244 kW. The next available chiller size was then selected from Carrier technical datasheet (Figure 6).

The model used for the analysis is the Carrier AquaForce - Fixed-speed water-cooled screw liquid chiller 30XW / 30XW-P 1052, which has a nominal capacity of 1327 kW and a COP of 5.45. This kind of chiller was chosen because it meets the technical requirements of the project. In other words, the cooling fluid used is not pure water only, but it is a kind of brine mixture which is able to operate at lower temperatures than water to avoid solidification.

The system works with exiting fluid temperatures of less than 5 °C and entering fluid temperatures greater than 10 °C. Therefore, it was very important that the water-cooled chiller selected be brine compatible. The Carrier unit selected can work in exactly these conditions, as demonstrated in the technical specification available in the previously mentioned datasheet. The selected power corresponds to approximately 377 refrigeration tons.

To estimate the cost per ton, market price references found online were used (Figure 7). Carrier AquaForce falls under the category of water-cooled screw chillers, which is priced at 600€ per ton. Since the selected capacity is above the indicated range (130-190 tons) but at the same time within the range for centrifugal chillers (150-3000 tons, at 350€ per ton), an estimated average value of 475€ per ton was considered.

Applying the above-mentioned factors, the estimated investment cost for a single chiller is around 179,000€, but since the system is designed with two chillers operating in parallel to provide more flexibility, the total CAPEX is estimated at 358,000€.

30XW--/30XWH-				852	1002	1052	1154	1252	1352	1452	1552	1652	1702
Heating													
Standard unit Full load performances*	HW1	Nominal capacity	kW	981	1185	1237	1324	1457	1557	1689	1795	1913	2001
		COP	kW/kW	5.98	5.77	5.67	5.79	6.12	5.96	5.76	5.61	5.94	5.92
	HW2	Nominal capacity	kW	958	1123	1174	1297	1375	1466	1592	1687	1867	1948
		COP	kW/kW	4.60	4.40	4.33	4.46	4.63	4.53	4.41	4.33	4.61	4.64
Seasonal energy efficiency**	HW1	SCOP _{30/35°C}	kWh/kWh	6.33	6.43	6.24	6.30	6.56	6.33	6.22	6.11	6.46	6.50
		η _{s heat 30/35°C}	%	245	249	242	244	254	245	241	236	251	252
		P _{rated}	kW	1153	1411	1473	1569	1737	1856	2013	2140	2265	2371
Cooling													
Standard unit Full load performances*	CW1	Nominal capacity	kW	829	1005	1049	1128	1242	1327	1438	1532	1637	1712
		EER	kW/kW	5.33	5.19	5.12	5.25	5.55	5.45	5.31	5.24	5.54	5.55
	CW2	Nominal capacity	kW	828	1188	1322	1220	1535	1677	1753	1865	1726	1830
		EER	kW/kW	5.43	6.93	6.30	5.75	6.72	6.71	6.30	6.36	5.95	5.91
Seasonal energy efficiency**		SEER _{12/7°C} Comfort low temp.	kWh/kWh	7.091	7.07	7.02	6.96	7.51	7.24	7.11	7.13	7.55	7.69
		η _{s cool 12/7°C}	%	281	280	278	275	298	287	282	282	299	304
		SEPR _{12/7°C} Process high temp.	kWh/kWh	8.01	8.29	8.11	7.96	8.97	9.09	8.34	8.13	8.45	8.50
Integrated Part Load Value	IPLV _{SI}		kW/kW	7.289	7.478	7.367	7.435	7.804	7.725	7.666	7.504	8.000	8.020

Figure 6: Carrier AquaForce technical data [6]

WATER COOLED CHILLERS	PRICE
Water Cooled Scroll Chiller 30-200 Tons	\$500/ton
Water Cooled Screw Chiller 130-190 Tons	\$600/ton
Water Cooled Centrifugal Chiller 150-3000 Tons	\$350/ton
Water Cooled Magnetic Bearing Chiller 150-550 Tons	\$700/ton

Figure 7: Prices of water-cooled chillers [7]

7 Scenario 2: Storage with Constant Chiller Production

After calculating the weekly expenditure of the base case, the project analyses the introduction of a storage system into the industry. In this scenario the storage system selected is an Ice Thermal Storage System. The goal with this scenario would be to cover the cooling demand during the whole week by a constant chiller production. In this scenario the first step was to calculate the average of the cooling demand, which is 1000.25 kWh.

The analysis began by determining the amount of cooling energy that could be either stored or discharged at each hour, depending on whether the actual cooling demand was below or above the average hourly demand. In periods where the demand was below the average, the energy in excess was considered as accumulated in the storage, while in the periods where the demand was above the average, the energy was required to be discharged from the storage system to meet the shortfall. From this hourly deviation, the accumulated energy was calculated as the cumulative sum of the differences between the stored and discharged energy over time. This metric reflects the net energy content within the storage system at any given hour and is critical for the sizing of the storage.

The primary objective was to adjust the reference average demand line to minimize the required storage capacity, defined as the maximum value reached by the accumulated energy curve. However, when proceeding this way a main issue was found: the accumulated energy was not always remaining non-negative. In order to solve this problem, the straight line of production was increased by an “objective value”. This way it was ensured that the storage system was never depleted to the point of being unable to supply the necessary cooling demand.

Using the function “Goal Seek” on Excel, it was then possible to find the lowest objective value that would generate no negative values in the storage state of charge through an iteration process. This value was found to be 15.34 kWh, meaning that the constant cooling production of the chiller is 1015.59 kWh. As the objective value, which is the value that lowers or heightens the constant cooling production line, gets lower, the minimum capacity of the storage lowers too. Therefore, an optimal balance must be identified, where the storage capacity is minimized without compromising the system’s ability to meet the cooling demand at all times.

To address the issue of accumulated energy in the storage system over multiple weeks, particularly due to the use of a constant baseline above the average demand to prevent negative storage levels, two approaches were considered:

- **Option 1 - Two-Week Balancing Cycle:** This method involves creating a two-week operational cycle to balance out the energy surplus. At the end of the first week, a positive energy balance is expected in the storage system, as the baseline is intentionally set above the average to avoid negative values. To compensate for this surplus, the baseline for the second week is lowered by an equivalent amount. This adjustment ensures that the cumulative energy over the two-week period returns to zero, effectively resetting the system at the end of each cycle. This method ensures balance over time but requires consistent and predictable demand patterns across weeks.
- **Option 2 - End-of-Week Adjustment (Selected Approach):** The approach implemented in the analysis simplifies the balancing process by ensuring the accumulated energy in the storage returns to zero by the end of the first week. Rather than extending the cycle into a second week, this method introduces a targeted adjustment to the baseline during the final hour of the first week. Specifically, the baseline for that hour is lowered by the exact amount of surplus energy accumulated during the week. This correction ensures that the storage system is fully discharged at the end of the week, maintaining a clean transition to the following period and avoiding energy carryover.

The second approach was chosen due to its simplicity and effectiveness in maintaining weekly energy balance without requiring multi-week modeling or assumptions about future demand behavior. In Figure 8, the final graph configuration is shown.

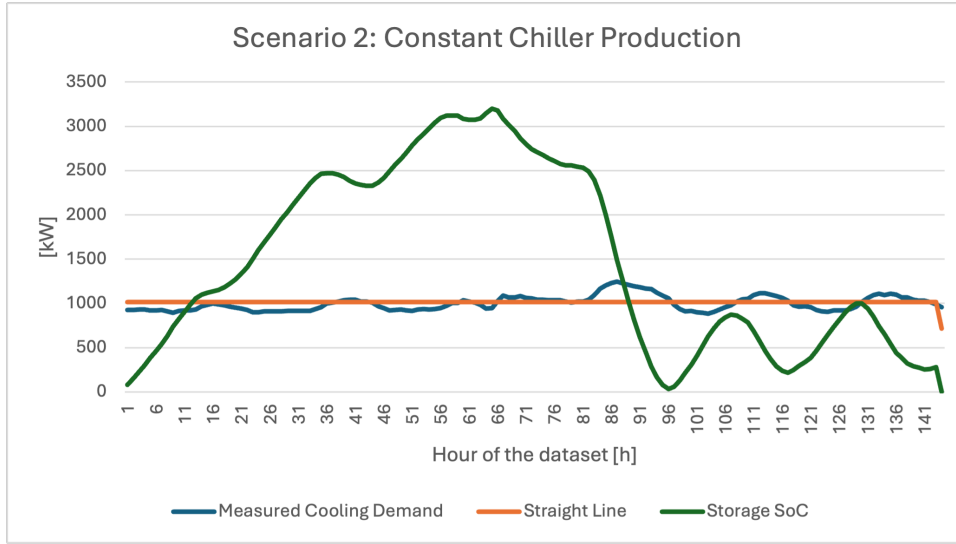


Figure 8: Constant chiller production configuration

In all the calculations made regarding this scenario, the losses related to the efficiencies that are considered are:

- Heat Exchanger Losses: an efficiency of 80% has been considered for heat exchanges in each transition from and to the storage system [8]
- the longer the fluid stays into the storage the more it will lose heat. In the model a loss of 0.3% each hour has been considered. This number has been found by calculations based on the thickness, area, material, storage fluid and other factors of the energy storage system.

From the calculation it has also been possible to find that the minimum capacity of the storage must be of 3198.8 kWh. To calculate the capital cost of the ice thermal energy storage then, a case study in Riyadh was analyzed [9]. It was possible to find the specific price in € per kWh stored, given the installed capacity and the CAPEX. Finally, the final formula used in the project is the following:

$$CAPEX_{\text{storage}} = \text{Minimum Capacity [kWh]} \cdot 45 \left[\frac{\text{€}}{\text{kWh}} \right]$$

In this case, being the capacity 3198.8 kWh, the CAPEX for the storage resulted to be 143,946.15€. From this scenario it was possible to find that the total cost of a constant production configuration is up to 853.61 € per week. It is possible to see that this cost is a little bit higher than the one found in the first scenario, and this can be correlated to the fact that in this scenario the losses related to the two-heat exchanger and the storage system were considered.

Having the storage capacity, it is possible to size the storage. Knowing that the latent heat of water (334 kJ/kg) that the storage should store 3198.8 kWh of thermal energy, the storage dimensions should fit 37.6 m³ of ice. Oversizing this storage by 15% to accommodate for the heat exchanger, the dimensions of the would be 4 meters high and 1.86 m radius. These values have been found using the volume equation of a cylinder, $V = \pi r^2 h$, and a density of ice of 917 kg/m³, since ice is used as PCM.

The sizing of the heat exchanger was done using the following equation:

$$A = Q / (U \Delta T_{LMTD}),$$

where A is the heat transfer area of the exchanger (in m²), Q is the rate of heat transfer (in W), U is the overall heat transfer coefficient (in W/m²·K), ΔT_{LMTD} is the log mean temperature difference (in °C). ΔT_{LMTD} was found to be 7.5465°C estimating that the temperature in the primary (cooling) loop is constant at 0°, and the temperature of the secondary (cooled) loop goes from 4.58°C (minimum supply temperature) to 11.58°C (maximum supply temperature). Q was assumed to be the maximal amount of heat being extracted by the storage in 1 hour, or 244.5 kW. U was assumed to be 200 W/m²·K. The size of the heat exchanger was found to be 162 m². Stainless steel tubing with a 22 mm outer diameter was selected as the heat exchanger material due to its high thermal conductivity, corrosion resistance, and commercial availability.

With these values, we can calculate the space needed for the heat exchanger:

$$V = A_{\text{heatexchanger}} * r_{\text{tubing}} = 162 \text{ m}^2 * 0.011 \text{ m} = 1.8 \text{ m}^3.$$

This can comfortably fit within the oversized storage designed. Implementing many smaller coils can help make the heat exchanger fit within the storage.

The storage insulation will be done using Polyurethane Foam, as it is a widely used insulator in commercial and industrial application and has a low thermal conductivity of 0.023 - 0.024 W/mK when it is waterproof coated [10]. Even if polyurethane foam with waterproof coatings is more expensive than regular polyurethane foam, it is useful for this application, as it will be exposed to a lot of moisture. Polyurethane foam with waterproof coatings is indeed very moisture resistant. Other alternatives such as extruded polystyrene have been considered but were later discarded. The storage tank will be made out of steel to minimize the risk of leakage. In order to reduce thermal losses, the storage tank will be put underground.

8 Scenario 3: Storage with Variable Chiller Production

In this third and final scenario, the chiller adapts its production based on the electricity prices of each hour.

The idea is to increase cooling production when electricity is cheap and reduce or stop it when prices are high, using the storage system to supply the demand during those more expensive hours. This method allows the system to reduce operational costs through what is known as price arbitrage. The problem of finding the optimal production profile in this case is quite complex and could not be fully minimized with standard tools. However, a locally optimal solution was found by setting specific conditions. As done in Scenario 2, the amount of energy in the storage at the last hour is set to zero, and that the chiller operates at maximum capacity at the first hour.

These conditions were necessary to maintain technical feasibility and to allow a proper cost evaluation over time, such as extending the results to a 15-year period. To determine how much the chiller should produce each hour, a price-dependent reduction factor table was introduced. Table 3 assigns a multiplier (ranging from 0 to 1) to the chiller's maximum capacity depending on the electricity price. For example, when the price is below 10€/MWh, the chiller works at full capacity. Between 10 and 45€/MWh, production is gradually reduced, while above 45€/MWh, the chiller is fully turned off.

The full list of reduction factors used in the final configuration is shown in Table 3. This approach resulted in a need for two chillers, each with a nominal cooling capacity of 2500 kW, and a storage system with a minimum capacity of about 22,570.8 kWh.

Table 3: Reduction factor as a function of electricity price threshold

Price below [€/MWh]	Reduction Factor
5	1
10	1
15	0.9
20	0.8
25	0.8
30	0.7
35	0.6
40	0.4
45	0.3
50	0
55	0
60	0
65	0
70	0
75	0
80	0

The system was therefore dimensioned to both absorb enough energy when prices are low and supply it when prices are high, without compromising the cooling demand. The dimensions of the cylindrical storage system

capable of storing 22,570.8 kWh will be 7 meters high and a radius of 3.73 meters (using the same methodology described in Scenario 2). The size of the heat exchanger needed was calculated using the same methodology described in Scenario 2. It was found that the maximal heat exchange rate was 1324.6 kW. The size of the heat exchanger should thus be at least 877.6 m². Such a heat exchanger would take up 9.65 m³, comfortably fitting within the oversized margin.

This is a very large storage, but is most probably oversized, due to the fact that the cooling demand for May 1st to May 6th has been assumed constant throughout the whole year.

In Figure 9, is possible to see how the chiller changes its production based on the electricity price. When the prices are very high the production is restricted to zero, whereas when the prices are very low the chiller are designed to express their maximum power output. All the assumptions and technical constraints used in the previous scenarios, such as storage being empty at the end of the period, efficiency losses in heat exchangers, and realistic sizing, were also applied here to maintain rigorous and technically feasible conditions.

From an economic point of view, this scenario shows a significant improvement for the weekly cost for the electricity consumption. However, when considering the total cost over fifteen years the total is up to 2,185,000€. This includes a CAPEX of 588,330€ for the two chillers and 1,015,685€ for the storage system. The operating cost during the observed period was 631€, and when extended to fifteen years, it amounts to 386,800€.

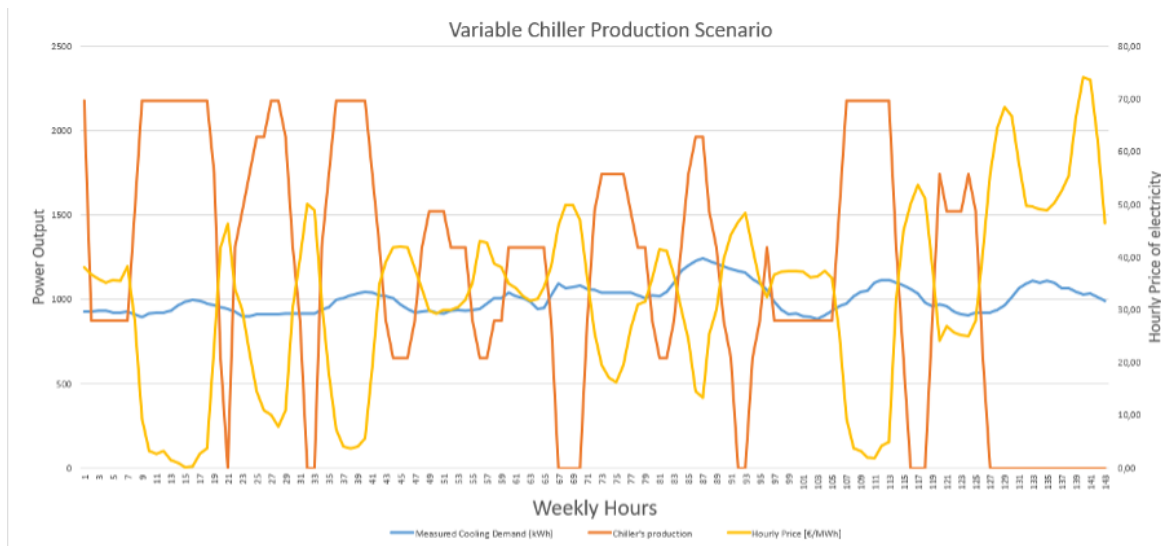


Figure 9: Variable Chiller Production configuration

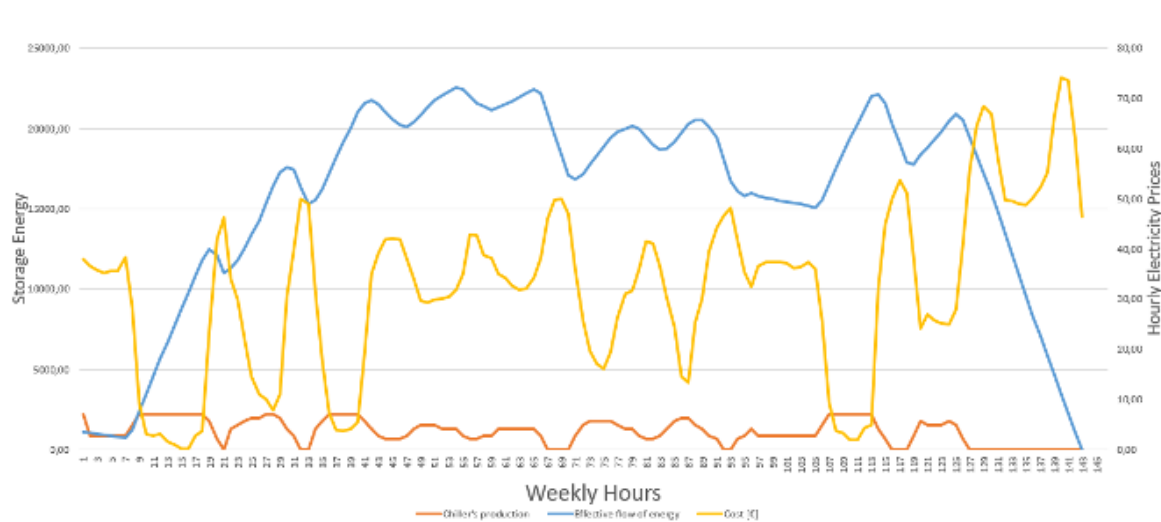


Figure 10: State of Charge of the Storage

9 Interpretation of the results

The economic analysis across the three scenarios reveals that the simplest configuration, operation without any storage, obtains the lowest total cost over both the one week and 15 year horizons. Specifically, Scenario 1 achieves a weekly cost of approximately 838.81€ and a 15 year total of around 1,133,000€, outperforming both storage enabled cases.

Despite its overall cost advantage, Scenario 1 does not exploit the potential for operational savings that arise from shifting energy consumption to periods of lower electricity prices.

Scenario 2 (constant chiller production with storage) incurs additional costs, up to 853.61 € per week, primarily due to heat exchanger inefficiencies (80% assumed) and hourly storage losses (0.3% per hour). The required storage capacity in this case is relatively modest (3,199 kWh), but the incurred losses outweigh any benefit from smoothing production.

Scenario 3 (variable chiller production with price driven charge/discharge) demonstrates the greatest reduction in operating expenditure, as the system charges storage during low price hours and discharges during high price hours. However, these OPEX savings are counterbalanced by significantly higher capital costs: two 2,500 kW chillers (588,330€) and a large storage tank (22,570 kWh, 1,015,600 €), resulting in a fifteen year cost of about 2,184,000€.

Nevertheless, the analysis of the three scenarios leads to several important insights:

- The most notable is that operational expenditure (OPEX) is significantly reduced in the third scenario, as it can be seen in Figure 11, indicating that incorporating a storage system can offer economic benefits under certain conditions.

		Scenario 1 - No Storage	Scenario 2 - Constant production	Scenario 3 - Variable Production
Chiller	Nominal Power [kW]	1,327.00	1,015.59	2,177.47
	Total Output [kWh/week]	143,035.66	145,943.50	161,567.99
	CAPEX Cost [€]	358,546.64	274,406.19	588,336.98
Storage	Minimum Capacity [kWh]	-	3,198.80	22,570.81
	CAPEX Cost [€]	-	143,946.15	1,015,686.57
Costs	Total CAPEX [€]	358,546.64	418,352.34	1,604,023.54
	Weekly OPEX [€]	843.79	853.61	631.42
	Lifetime OPEX [€]	775,346.72	784,363.27	580,202.14
	Total [€]	1,133,893.36	1,202,715.61	2,184,225.69

Figure 11: Summary of the three scenarios

- The second key observation is that the selected industrial case is characterized by a remarkably stable cooling demand compared to the other case studies. As illustrated in the figures below (e.g., Figures 12, 13, 14), both the office and shopping center scenarios exhibit significant fluctuations in cooling demand throughout the day. In those cases, implementing a storage system that enables a steady, linear operation of the chillers could yield substantial benefits by avoiding frequent ramping up and down, which is both energy-intensive and costly. In contrast, due to the relatively uniform demand pattern in the industrial case, there is little potential for operational smoothing.

As a result, the analysis focused primarily on the economic opportunity offered by electricity price fluctuations. Rather than reducing chiller peak load or sizing, the storage system in this study was evaluated for its potential to shift energy use to lower-cost hours.

In the shopping center example, for instance, the storage system clearly allows for the use of a significantly smaller chiller. The maximum cooling output required drops from approximately 3,500 kW without storage to around 1,400 kW with storage, effectively halving the required chiller capacity and thus reducing capital and operational costs.

In contrast, for the industrial scenario under the constant production setup, the chiller's peak output only

decreases from 1,327 kW to 1,015 kW, an appreciably smaller reduction, highlighting the limited potential for capacity downsizing in such a stable demand context.

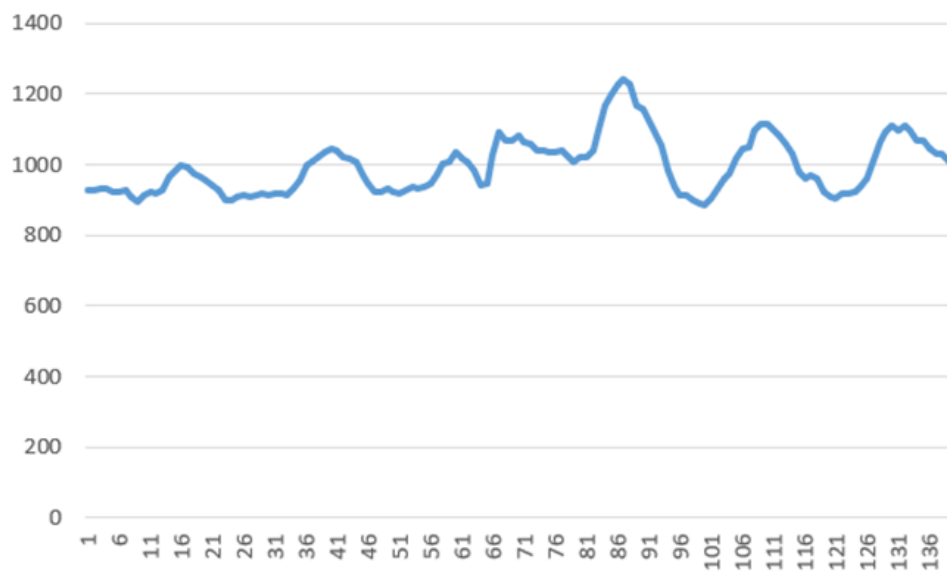


Figure 12: Industry weekly demand

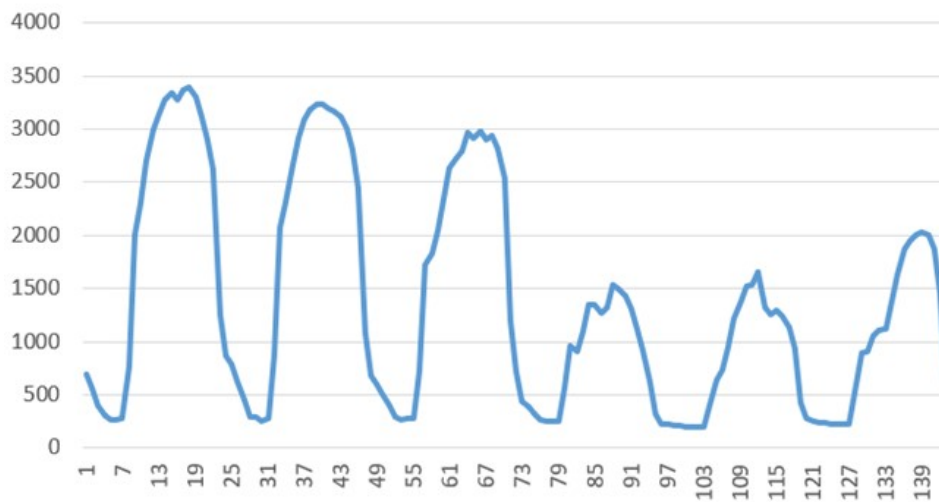


Figure 13: Shopping center weekly demand

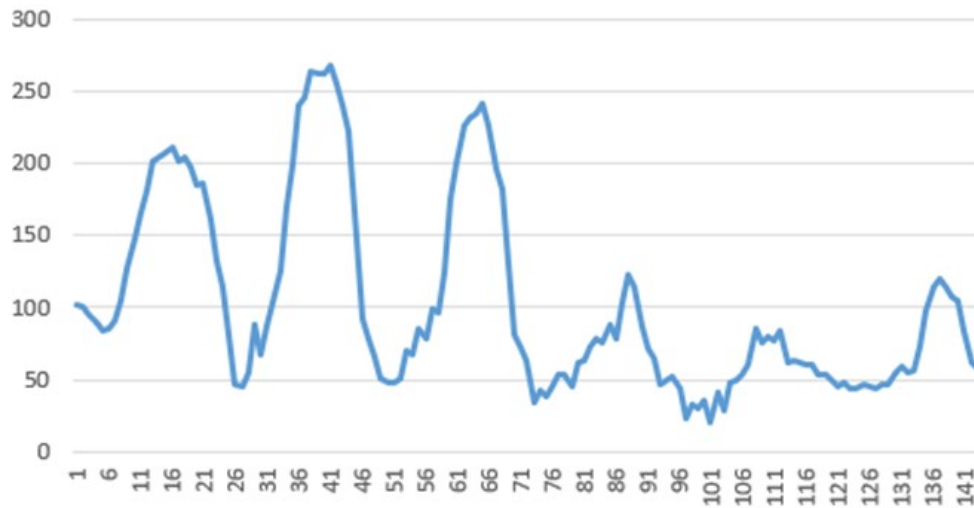


Figure 14: Office weekly demand

- Thirdly, the Excel model has some clear problems. In both scenario 2 and scenario 3, the battery's charge level looks like a "hill" (see Fig. 15): it fills up in the early hours of the data and then empties completely by the end. This happens because the chiller's output doesn't change based on how full the battery is.

In the constant-production case, the chiller runs at a steady rate so that the battery never goes empty, keeping the solution valid but means the model can't adapt to changing needs. In the variable-production case, the chiller only cares about the hourly electricity price. Also, the battery has to end the period at the same level it started, and the only way to make that work is to force it down to zero at the end, which keeps the required battery size as small as possible in this setup.

It were also tried Excel add-ins like Solver and OpenSolver for the third scenario, but none of them were able to find a solution.

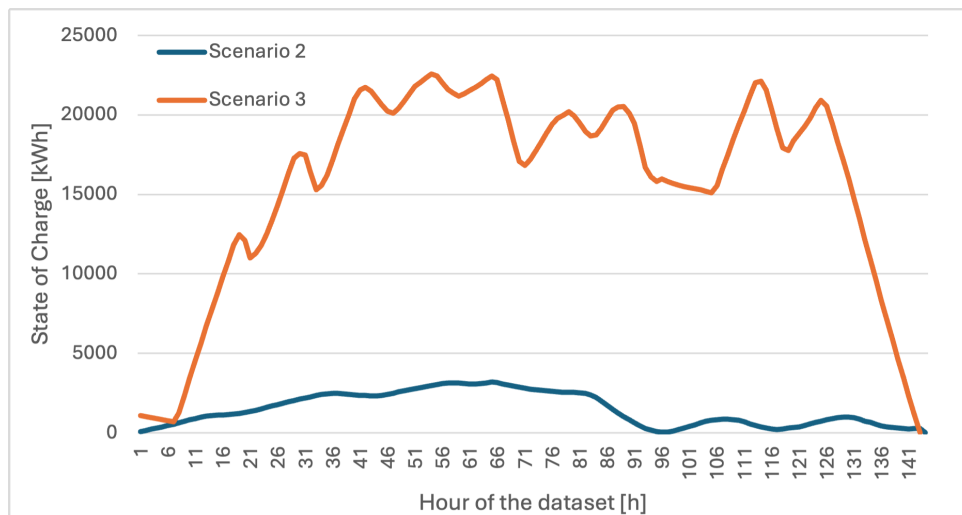


Figure 15: Storage State of Charge in the 2nd and 3rd scenarios

- Another important consideration is represented by the strong dependence of the storage system's capacity on the conditions at the beginning and end of the week. As illustrated in the following graph, if only the central days of the week are considered, the minimum required storage capacity would be significantly lower than the observed value of 22,570 kWh. The purple line in the figure below represents the hypothetical zero state of charge under this reduced timeframe, highlighting the impact that the boundary days have on the overall system behavior. This effect is caused by the substantial variation in electricity prices across the week. On the first day, prices are particularly low, leading the model to charge the storage system in a large way. In contrast, the last day is characterized by very high prices, prompting the model to discharge stored energy to maximize economic return. This pricing asymmetry drives a storage strategy that results in a higher calculated capacity than it would have been necessary under more constant pricing conditions.

These findings underscore the sensitivity of the storage model to short-term price dynamics and dataset boundaries. Consequently, reliance on a single week of data for evaluating long-term storage requirements or system performance may lead to inaccurate conclusions. A more comprehensive dataset accounting for seasonal trends and variability in both consumption and pricing, is essential for reliable long-term assessment and system design.

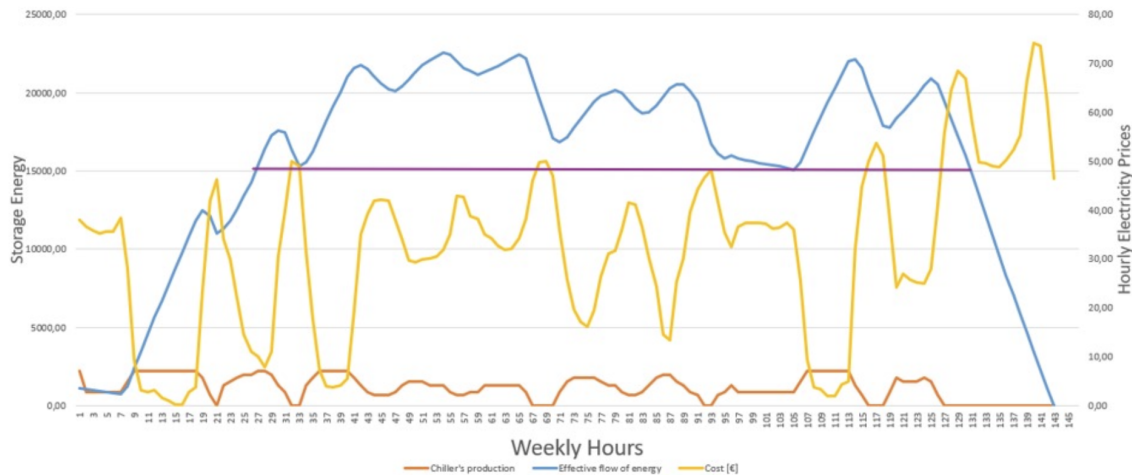


Figure 16: SOC of the third scenario with an hypothetical zero state of charge

Decision makers should therefore estimate the trade off between a high starting capital commitment and downstream savings: if budget flexibility allows for higher initial investment, a dynamically dispatched storage system can deliver continuous OPEX reductions and mitigate exposure to high market prices. Conversely, for operations constrained by capital availability or seeking simplicity, the direct chiller to load baseline remains the least costly option.

10 Conclusion

The selected industrial site exhibits minimal variation in electricity demand throughout the week, in contrast to applications such as office buildings, where significant fluctuations in load are common. This low variability diminishes the conventional value of an energy storage system, which typically lies in its capacity to absorb excess energy during periods of low demand and release it during peak loads. In this case, the stable energy consumption pattern results in a consistently distributed load across the day, thereby eliminating the need for storage to buffer against sudden ramp-ups or ramp-downs in chiller operation. Due to the absence of pronounced demand peaks, the role of thermal storage in smoothing out chiller load becomes marginal. Consequently, the focus of the analysis shifts from operational smoothing to economic optimization, specifically, leveraging temporal differences in electricity prices. The intent is to charge the storage system when electricity is inexpensive and discharge during high-cost periods, thereby reducing overall operating expenditure through price arbitrage.

The analysis highlights a key limitation in the current modelling approach, which is primarily based on electricity price signals rather than on demand-side dynamics such as peak shaving or load shifting. Given the inherent volatility and unpredictability of electricity prices, using only a single week of data proves insufficient for making reliable long-term assessments of storage system performance or sizing. This issue is further compounded by the characteristics of the specific industrial case under study, where electricity demand remains relatively stable over time. Unlike other case studies where demand fluctuates significantly, ranging, for example, from 3,500 kW to as low as 400 kW, the low variability in this context eliminates many of the potential operational benefits of a storage system.

As a result, the interaction between a highly price-sensitive model and a flat demand profile amplifies the limitations of using short-term data and leads to conclusions that may not generalize or reflect actual system needs over longer periods. In this particular scenario, the justification for implementing a storage system becomes significantly weaker, as the primary value driver, price arbitrage, cannot be confidently captured or forecasted with such limited input data. This underscores the need for broader datasets and more comprehensive modeling frameworks that

incorporate both market volatility and demand-side flexibility to more accurately evaluate the feasibility and effectiveness of energy storage solutions.

To conclude, results indicate that for facilities with consistently uniform cooling loads, the benefits of thermal storage are limited. The introduction of storage systems leads to additional energy losses and substantial capital expenditures, which cannot be offset solely through reduced cycling or peak shaving. Although Scenario 3 demonstrates a reduction in operating costs via strategic energy shifting, these savings are outweighed by the increased capital investment required, particularly for larger storage capacities and dual-chiller configurations. Under the current market conditions and with the observed demand characteristics, the most cost-effective approach remains the direct operation of chillers without any storage integration. Future studies should explore hybrid strategies that incorporate small-scale storage combined with dynamic chiller controls, assess the feasibility of alternative storage technologies with reduced losses, and evaluate the potential of time-of-use pricing models to determine the economic tipping point for storage deployment.

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